APS independent public-fixture replay

Native implementation follow-up

29 October 2026

Question and boundary. The earlier APS instrument checked results with its own fast curve arithmetic. This continuation adds a separate native replay path for its three existing public, known-answer fixtures. It checks the mathematical transcript and sampling, while explicitly declining to attest execution history or solver cost counters. No performance experiment, scaling claim or new curve family is introduced.

Method. Strict parsing rejects unknown fields, duplicate or aliased coordinate-map keys, unsupported metadata and oversized inputs. The checker binds the fixture model, generator, target, public answer and ICV1 identity. It uses existing BigUint field and point addition/doubling, with affine double-and-add for known scalars; it never calls the producer's FastCurve, sampler, polynomial solver or recovery routines.

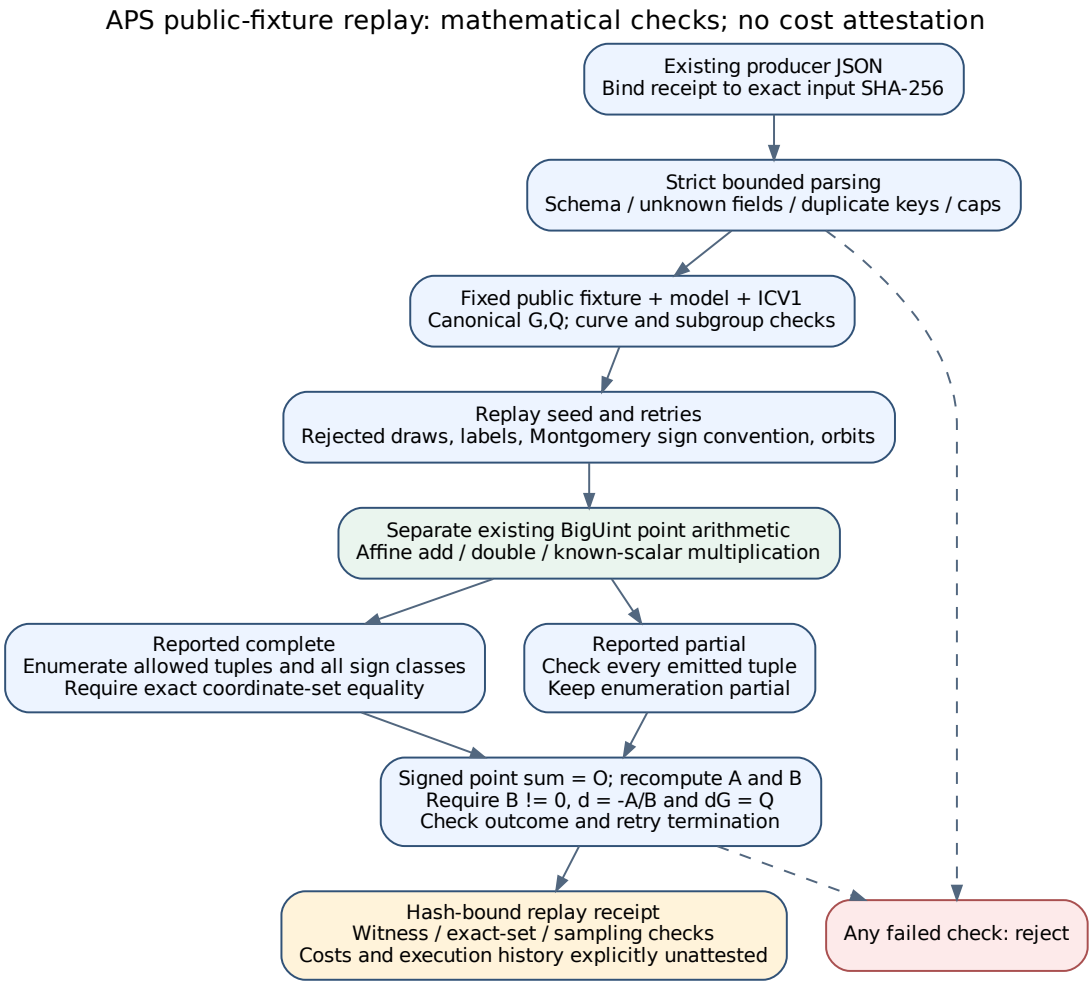
For each attempt it regenerates the seeded label draws, discarded infinities and repeated abscissae, partitions, sign representatives and orbit expansion. The sign representative compares the Montgomery encoding of y to $(p-1)/2$; the verifier reproduces that convention with BigUint rather than assuming the affine integer y is the smaller lift. Every recorded point must satisfy its independent label equation $R_i = u_i G + v_i Q$.

For every reported coordinate tuple the verifier enumerates signs and tests the actual point sum. A tuple must satisfy the arm's partition or block membership. A pruned line arm additionally excludes horizontal triples. For a complete attempt it independently enumerates all allowed distinct tuples and requires equality with the reported solution set. For a partial attempt it validates the emitted subset and preserves partial status.

A recovery must bind to a reported tuple and its canonical sign orientation. The checker recomputes $A = \sum s_i u_i$ and $B = \sum s_i v_i$, requires $\sum s_i R_i = O$ and $B \neq 0$, computes $d = -A/B \pmod{n}$, and checks $dG = Q$. The public expected answer alone cannot authorize a relation: a horizontal $B = 0$ orbit remains a rejection even when its supplied d is correct.

Why the finite oracle covers the sign choices. Every allowed tuple has three or four distinct abscissae. Choosing one fixed lift per abscissa leaves only its sign to select. Simultaneously negating all signs preserves the zero-sum condition, zero/nonzero status of B, and recovered quotient. Thus fixing the first sign positive covers every coordinate decomposition. This applies the established group law; it is not a new mathematical result.

Validation. Final command receipts, failure history and source digests are maintained in the study README and validation directory. The tests cover independent sampling, fixture binding, complete and partial sets, status/retry consistency, corrupted labels and witnesses, duplicate JSON keys, resource bounds and every CLI arm in both enumeration modes. Only completed checks in those receipts count as passed.



Scope of independence. EC arithmetic and relation checking use a separate implementation. Serialization, rand 0.8 StdRng and the ICV1 metadata convention are shared dependencies. The byte hash binds the receipt to the input, not to a physical execution or a formal proof. The existing partial Lean coefficient proofs remain unchanged; no new Lean/Rust correspondence claim is made.

Remaining work. Full operation accounting, native ecbench/export integration, metric validation, frozen controlled inputs and matched external solver/rho comparisons remain open. Aborts before a v1 transcript is emitted are not covered by this replay format. Prior rejected source-checkpoint and shared-file uploads remain on hold; no performance or EKS job is launched. The IC leaderboard, performance charts and canonical curve graphs receive no new values: this work establishes no measured gain or new curve route.

Sources. SEC 1 v2.0, sections 2.2.1 and 3.2.2.1, for the group law and point validation; `src/ecc/point.rs` and `src/ecc/field.rs`, the reused arithmetic; `src/cryptanalysis/aps_single_relation/replay.rs` and `examples/aps_single_relation_check.rs`, implementation and controls. The earlier APS source map and four-lead study preserve the manuscript, chord-law references, negative results and formalization boundary.